

git commands step by step

1. initialize a Git repository:

git init

2. Start tracking changes to your files by adding them to the staging area:

staging : means changes to files are marked for inclusion in the next commit. When you make changes to your files in a Git repository, Git doesn't automatically track or commit those changes. Instead, you have to explicitly tell Git which changes you want to include in the next commit by staging them.

git add <file1> <file2> ...

You can also add all files at once using:

git add .

3. Commit Changes □ save changes

git commit -m "Your commit message here"

4. **Check Status:** To see the status of your working directory and staging area:

git status <https://www.javatpoint.com/git-status>

it provides information about the current state of your working directory.

like:

- Status when there are no changes made:

On branch master

nothing to commit, working tree clean

- Status when a new file is created

create a file using the **touch** command

\$ touch file_name

\$ git status

On branch master

Untracked files:

file_name

nothing added to commit but untracked files present (use "git add" to track)

- after staging the file

changes to be committed

new file: file_name

5. View Commit History:

git log

it is used to read a history of everything that happens to a repository. or it is a record of commits/changes you have done to your repo.

A git log contains the following data:

- **A commit hash**, which is a 40 character checksum data generated by SHA (Secure Hash Algorithm) algorithm. It is a unique number.
- **Commit Author metadata**: The information of authors such as author name and email.
- **Commit Date metadata**: It's a date timestamp for the time of the commit.
- **Commit title/message**: It is the overview of the commit given in the commit message.

6. Push Changes to Remote Repository:

git push origin <branch_name>

7. Pull Changes from Remote Repository: If there are new changes in the remote repository, you can pull them into your local repository:

git pull origin <branch_name>

8. Create a New Branch:

git branch <branch_name>

9. Switch to a Branch:

git checkout <branch_name>

10. Merge Branches:

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git checkout <branch_to_merge_into>
git merge <branch_to_merge_from>
```

11. Remove Files from Staging Area:

If you accidentally add files to the staging area and want to remove them:

git reset HEAD <file>

12. Discard Changes in Working Directory:

To discard changes in your working directory (be careful, this is irreversible):

git checkout -- <file>

13. **Remove Untracked Files:** from your working directory:

git clean -f

14. git merge header